Tennessee Technological University
Department of Computer Science

CSC 2400 — Design of Algorithms
Bryson Bargas, Noah Boggess

CVRP Algorithm Comparison — Checkpoint 2 Progress Report

1. Current Progress

At this stage of the project, we have transitioned from planning into implementation. We have successfully set up our GitHub repository with the required structure and begun developing the core algorithms for our comparison.

We are working with three algorithms:

- Greedy Nearest Neighbor (baseline)
- Clarke–Wright Savings Algorithm
- Local Search (possible improvement)

We removed the Ant Colony algorithm from the group (after previous approval) due to the individual choice of one of our group members to leave the team.

So far, the Greedy, Clarke–Wright algorithms, and Local Search are fully implemented and tested on "The Solomon Benchmark" from Professor Marius M. Solomon

2. Current File Design

We currently have four files, three hosting our previously mentioned algorithms, and one additional used for calling those algorithms and allowing selection of the data set to be tested against once.

- Runs each algorithm on the same input
- Measures runtime using Python's timing functions
- Computes total route distance as the primary measure of solution quality

3. Key Design Decisions

Several important decisions were made early in development:

- **Data Set Generation:**
  We decided to not generate data sets, which is what we started out doing, and instead started using data sets online (The Solomon Benchmark)

- **Algorithm Scope:**
  We chose to drop Ant Colony Optimization due to our group member leaving the project and instead focus on improving and comparing three core algorithms more thoroughly.
- **Local Search Approach:**
  We decided to implement a 2-opt based local search that improves routes generated by greedy. This allows us to clearly demonstrate improvement-based optimization.

4. Current Results and Observations

Initial testing shows expected behavior:

- The Greedy algorithm runs very quickly and consistently produces valid routes, but the total distance is relatively high.
- The Clarke–Wright algorithm takes slightly longer but generally produces shorter routes compared to greedy.
- Local Search is a slight improvement to Greedy algorithm, running very quickly and producing slightly less distance results.

These early results align with our expectations about the trade-off between execution time and solution quality.

5. Challenges Encountered

The main challenges so far have been:

- **Implementing Clarke–Wright merging logic**, especially ensuring capacity constraints are respected
- **Designing Local Search**, particularly deciding how to represent routes and apply improvements cleanly
- **Working with algorithm data,** all algorithms needed to be tested on identical datasets and evaluated using the same metrics, which proved to be a challenge.

We also had to ensure that all algorithms produce outputs in the same format so they can be compared directly. Also figuring out how to make the algorithms in general was difficult, however, we got there in the end.